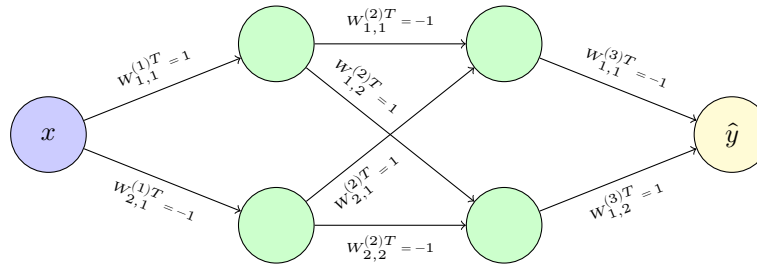


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1 Backpropagation

- What is the advantage of backpropagation vs. computing the gradients in isolation for every parameter ?
- You are given a network with 3 layers, no bias terms and with ReLU activation functions for the hidden layers (green). The weights are initialized as shown in the figure below, the learning rate is $\eta = 0.01$ and the mean squared error is used as a loss function. Given a dataset of only one example that has only one feature $X = \{3\}$ and label $y = \{6\}$, calculate the loss of the network and update the weights to their new values using backpropagation.



2 Convolutional Neural Network (CNN)

- What is the main structural difference between CNNs and Feed Forward Neural Networks?
- You are given a 4×4 grayscale image and a convolutional layer that features 2 kernels of size 2×2 with stride 1. What would be the output size after feeding the input image to the convolutional layer? What would be the output size if we increased the stride to a value of 2? With stride, we refer to the sliding movement of the kernel. Do not apply any padding.
- Assume that you have an image I and a kernel K :

$$I = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 5 & 3 \end{pmatrix} \quad K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (1)$$

Compute the convolution $O = I * K$ without using padding, i.e. the output matrix has dimensions $O \in \mathbb{R}^{2 \times 2}$.

3 Neural Networks and Regularization

L_P regularization penalizes the p -norm of the weights, i.e., $L_P = \|\mathbf{w}\|_p^p = \sum_i w_i^p$. This regularization generalizes L_1 and L_2 regularization. Compute the gradients of the weights for the loss function using L_P regularization:

$$\tilde{J} = J(\mathbf{w}) + \alpha \|\mathbf{w}\|_p^p \quad (2)$$

$$\nabla_{\mathbf{w}} \tilde{J} = ? \quad (3)$$